Lecture 10 Partial Derivatives

Continue ...

Example: Suppose that $w = x^2+y^2-z^2$ and $x = \rho \sin\phi \cos\theta$, $y = \rho \sin\phi \sin\theta$, $z = \rho \cos\phi$.

Using appropriate forms of the chain rule to find $\frac{\partial w}{\partial \rho}$ and $\frac{\partial w}{\partial \theta}$.

Sol^n: $$\frac{\partial w}{\partial \rho} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \rho} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \rho} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \rho} \quad \text{——— (1)}$$

$w = x^2+y^2-z^2$

$\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = -2z$

$x = \rho \sin\phi \cos\theta$, $y = \rho \sin\phi \sin\theta$, $z = \rho \cos\phi$

$\frac{\partial x}{\partial \rho} = \sin\phi \cos\theta$, $\frac{\partial y}{\partial \rho} = \sin\phi \sin\theta$, $\frac{\partial z}{\partial \rho} = \cos\phi$

From (1)
$$\frac{\partial w}{\partial \rho} = 2x \sin\phi \cos\theta + 2y \sin\phi \sin\theta - 2z \cos\phi$$
$$= 2(\rho \sin\phi \cos\theta) \sin\phi \cos\theta + 2(\rho \sin\phi \sin\theta) \sin\phi \sin\theta - 2(\rho \cos\phi) \cos\phi$$
$$= 2\rho \sin^2\phi \cos^2\theta + 2\rho \sin^2\phi \sin^2\theta - 2\rho \cos^2\phi$$
$$= 2\rho \sin^2\phi (\cos^2\theta + \sin^2\theta) - 2\rho \cos^2\phi$$
$$= 2\rho (\sin^2\phi - \cos^2\phi)$$
$$= -2\rho \cos 2\phi$$

$\boxed{\frac{\partial w}{\partial \theta} \text{ do yourself !!!}}$

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Implicit Differentiation

Theorem 1:

If the equation $f(x,y)=c$ defines y implicitly as a differentiable function of x, and if $\frac{\partial f}{\partial y} \neq 0$, then

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}$$

Example: Given that $x^3+y^2x-3=0$, find $\frac{dy}{dx}$.

Sol^n:

$$f(x,y) = x^3+y^2x-3$$

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} \quad \text{——— (1)}$$

$$\frac{\partial f}{\partial x} = 3x^2+y^2, \quad \frac{\partial f}{\partial y} = 2xy$$

So, $$\frac{dy}{dx} = -\frac{3x^2+y^2}{2xy}.$$

——— x ———

Let $z = f(x,y)$

$$* \quad \frac{dz}{dx} = \frac{\partial f}{\partial x}\frac{dx}{dx} + \frac{\partial f}{\partial y}\frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\cdot\frac{dy}{dx}.$$

if $z = f(x,y) = c$ then $\frac{dz}{dx} = 0$ so,

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\cdot\frac{dy}{dx} = 0 \Rightarrow \boxed{\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}}$$

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Theorem 2:

If the equation $f(x,y,z)=c$ defines z implicitly as a differentiable function of x and y, and if $\partial f/\partial z \neq 0$, then

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}.$$

Example: Consider the sphere $x^2+y^2+z^2=1$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the point $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$.

Sol^n.

$$f(x,y,z) = x^2+y^2+z^2-1$$

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial z} = -\frac{2x}{2z} = -\frac{x}{z}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z} = -\frac{2y}{2z} = -\frac{y}{z}$$

at the point $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$.

$$\left.\frac{\partial z}{\partial x}\right|_{(2/3,1/3,2/3)} = -\frac{2/3}{2/3} = -1$$

$$\left.\frac{\partial z}{\partial y}\right|_{(2/3,1/3,2/3)} = -\frac{1/3}{2/3} = -\frac{1}{2}$$

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DO yourself:

1. Find $\frac{dz}{dx}$ for $z = x\ln(xy) + y^3$, $y = \cos(x^2+1)$

2. Use the chain rule to find $\frac{dz}{dt}$.

   (a) $z = 3x^2y^3$; $x = t^4$, $y = t^3$

   (b) $z = 5\cos xy - \sin x$; $x = \frac{1}{t}$, $y = t$, $q = t^2$

   (c) $z = \sqrt{1+x-2yq^4x}$; $x = \ln t$, $y = t$, $q = 3t$

3. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ form the ~~bellow~~ equations:

   (a) $x^2 - 3yz^2 + xyz - 2 = 0$

   (b) $ye^x - 5\sin 3z = 4z$

4. Let $w = 4x^2 + 4y^2 + z^2$, $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, $z = \rho\cos\phi$.
Using chain rule to find $\frac{\partial w}{\partial \rho}$, $\frac{\partial w}{\partial \phi}$ and $\frac{\partial w}{\partial \theta}$.

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